

Match the algebraic statement to its numeric example.

1.

Property	Numeric Example
Division Property of Equality	$4 + 5 + 7 = 9 + 7$
Subtraction Property of Equality	$(4 + 5) \times 7 = 9 \times 7$
Multiplication Property of Equality	$(4 + 5) \div 7 = 9 \div 7$
Symmetric Property of Equality	$4 + 5 = 9$
Addition Property of Equality	$4 + 5 - 7 = 9 - 7$

Consider the equation $33t - 99 = 0$ to answer questions 2-5.

- Complete the statement to describe the equation. The variable, t , is first being _____ by 33, and then 99 is being _____ from this product. The value of the expression, $33t - 99$ is _____ to 0.
- Describe the operation that could be performed to isolate $33t$ on the left side of the equation.
- Describe the operation that could be performed to isolate the t on the left side of the equation.
- Solve $-4 + \frac{x}{2} = 16$ by applying properties of equality. Explain which properties were used to solve the equation.

6. Solve $12n + 9 = 93$ by applying properties of equality. Explain which properties were used to solve the equation.

Use the equation $6x - 12 = 30$ to complete the following.

7. Apply properties of equality to solve the equation. Use substitution to verify the solution.

Use the equation $5y + 7 = 42$ to complete the following.

8. Apply properties of equality to solve the equation. Use substitution to verify the solution.

Use the equation $\frac{x}{5} - 8 = 17$ to complete the following.

9. Apply properties of equality to solve the equation. Use substitution to verify the solution.

Use the equation $\frac{y}{5.2} + 3.4 = 4.3$ to complete the following.

10. Apply properties of equality to solve the equation. Use substitution to verify the solution.